

GPA COST CONTROL ERP

Commercial ERP + HRIS Product Roadmap

From internal tool → market-ready construction ERP

"One system for running a construction company —
from site costs to payroll."

	Target Market	Construction & project-based companies	On-premise license + Implementation fee
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6 months
Revenue

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1. Current ERP State — Readiness Assessment

The GPA Cost Control ERP V5.0 was built and battle-tested at PT Garuda. The ERP core is ~90% production-ready. HRIS is the major greenfield. Below is the current readiness per module.

Auth & RBAC	100%	JWT, 7 roles, menu permissions, full
Action Center	95%	Role-based queues, batch ops
Project Command	95%	CRUD, import Excel, budget bar, health dots
Revenue (AR)	95%	Full lifecycle, confirmed AR drives budget ceiling
Spending / Expenses	95%	Full workflow, receipt upload, voucher print
Petty Cash	90%	Batch entry, auto-creates draft expenses
Inventory & Assets	90%	Full CRUD + transaction log
Legal & Proposals	90%	Full workflow, MD signature upload
Notifications	85%	In-app bell with polling; email not yet
Settings	90%	Profile, password, user management done
Dashboard	85%	Functional; needs unified API endpoint
Reports	75%	Data renders; Excel export button not wired
HRIS — All modules	0%	Not started — primary build target

Key insight: The ERP engine is proven. The commercialization investment is primarily in HRIS (~42 developer-days) and the productization scaffolding (~9 days) that allows clean installation at a new client's server.

2. What Needs to Be Built

2.1 Productization Layer

These are not features — they are the scaffolding required to ship the product to a second client cleanly. Without this layer, every new install requires manual database patches and hardcoded config.

Company Config Table	1 day	Company name, logo, NPWP, address — used in all PDF headers and payslips. Each client customizes this via the admin panel after install.
Alembic Migrations	2 days	Replace the ad-hoc <code>_ensure_incremental_schema()</code> function with proper versioned Alembic migrations. Required before any second install — otherwise schema drift is unmanageable.
Installer Script	1 day	<code>setup.bat</code> / <code>setup.py</code> : creates virtualenv, installs dependencies, runs migrations, seeds roles and menus. One command turns a blank server into a running system.
First-Run Setup Wizard	1 day	On first launch with empty DB, redirect to <code>/setup</code> . Admin fills in company profile and creates the first <code>SUPER_ADMIN</code> account. No manual DB seeding needed.
Backup / Restore CLI	1 day	<code>python manage.py backup</code> → <code>pg_dump</code> to timestamped archive. <code>python manage.py restore</code> → safely restores. Critical for on-premise support contracts.
Environment Config Wizard	0.5 day	<code>.env</code> generator script: DB URL, <code>SECRET_KEY</code> , port, company slug. Forces required secrets — no more hardcoded fallbacks.
White-Label PDF Templates	1 day	All generated PDFs (expense voucher, payslips, legal docs) pull company name and logo from the config table instead of hardcoded strings.
Excel Export — Reports	1 day	Wire the existing Export button in the Reports page. Currently renders data but has no <code>onClick</code> handler. Blocks every sales demo.

Productization: ~9 developer-days

Complete in Month 1 alongside Garuda s

2.2 HRIS H1 — Employee & Organisation

The foundation of HRIS. Every other module (payroll, attendance, recruitment) depends on having accurate employee master data.

Backend: 5 days	Frontend: 5 days	Total: 10 days
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Employee master record: NIK, NPWP, BPJS TK no., BPJS Kesehatan no., bank account details

Departments & job grades — hierarchical org structure, links to approval matrix

Employment types: Permanent (PKWT), Contract (PKWTT), Outsource

Org chart: visual tree view, exportable to PDF

Document store: upload KTP, SKCK, ijazah, BPJS cards per employee

Link to ERP user: employee.user_id → users.id (optional — not all employees log in to the system)

Employment history: start date, promotion dates, termination with reason

Integration with ERP: Employee records link to the existing Users table for login. Department hierarchy informs the approval matrix — department heads can be auto-assigned as approvers for their team's expenses.

2.3 HRIS H2 — Attendance & Leave

Daily attendance tracking and leave management. Feeds directly into payroll calculation — absent days reduce base pay, overtime adds to it.

Backend: 5 days	Frontend: 5 days	Total: 10 days
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Daily attendance: manual entry or CSV import from fingerprint / face recognition machine

Overtime recording: regular hours, weekend rate (1.5×), holiday rate (2×) per Permenaker 2023

Leave types: Annual (cuti tahunan 12 days/yr), Sick, Personal, Maternity/Paternity, Unpaid

Leave balance engine: auto-accrual, configurable carry-over policy per company

Approval flow: Staff → direct Manager → HR — reuses the existing ERP approval pattern

Calendar UI: monthly view per employee + team calendar for managers

Absensi rekap: monthly summary exportable to Excel for payroll input

Integration with ERP: Attendance data feeds into the Payroll module (H3). Overtime hours and absent days are automatically pulled into the monthly payroll run, eliminating manual reconciliation between HR and Finance.

2.4 HRIS H3 — Payroll

The highest-value HRIS module. Indonesian payroll is complex — PPh 21, BPJS TK, BPJS Kesehatan, THR, and bank disbursement. Getting this right is the main technical risk and the main reason clients will pay for the system.

Backend: 8 days	Frontend: 6 days	Total: 14 days
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Salary component engine: Basic, transport, meal, positional, project allowances — fully configurable
BPJS Ketenagakerjaan: JKK + JKM (employer), JHT (2% employer + 3.7% employee), JP (2% + 1%)
BPJS Kesehatan: 4% employer + 1% employee, capped at BPJS ceiling
PPh 21: gross-up or netto method, PTKP per status (TK/0, K/0, K/1, K/2, K/3), progressive tariffs 2024
THR: 1/12 x months of service x basic salary, auto-calculated at Lebaran period
Payroll run workflow: Period lock → Calculate → Review → Approve → Post (with immutable audit log)
Payslip PDF: company header, employee details, earnings/deductions breakdown, net pay in terbilang
Bank disbursement CSV: BCA, Mandiri, BNI, BRI payroll file formats
Payroll rekap: summary by department and cost centre — links directly to ERP cost centre codes

Critical Risk: PPh 21 calculations must be validated by a certified accountant (Akuntan Publik) before shipping to any client. A miscalculation creates tax liability for the client. Budget 2-3 days for accountant review in Month 5.

Integration with ERP: Payroll cost posts to ERP expenses by cost centre. Salary disbursement = an approved Expense. Finance sees total labour cost broken down by project/department without double-entry.

2.5 HRIS H4 — Recruitment & Onboarding

Lowest technical priority — but rounds out the 'one system' sales pitch. Recruitment is often what HR managers ask about first.

Backend: 4 days	Frontend: 4 days	Total: 8 days
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Job postings: internal-only in v1 (no job board API integration)
Applicant tracking: Received → Screening → Interview → Offer → Hired/Rejected
Interview scheduling: date, interviewer assignment, notes and scoring
Offer letter: auto-generated PDF with salary, position, start date
Onboarding checklist: tasks assigned to HR and new hire, completion tracking
Auto-create employee: on 'Hired' status, pre-fills Employee master from applicant record

Integration with ERP: New hire's Employee record auto-creates a login User account and assigns the correct role. Onboarding tasks can include 'Complete expense training' and 'Submit first petty cash request' linked to the ERP workflows.

3. Six-Month Execution Timeline

Total build effort: ~51 developer-days across 6 months. Month 1 runs in parallel with Garuda stabilization.

	Month 1	Month 2	Month 3	Month 4	Month 5	Month 6
ERP Stabilization						
Productization Layer						
Excel Export + Fixes						
Alembic Migrations						
HRIS H1: Employee						
HRIS H2: Attendance						
HRIS H3: Payroll Pt1						
HRIS H3: Payroll Pt2						
HRIS H4: Recruitment						
Documentation						
Pilot Client Onboard						

Key Milestones

End of Month 1	Garuda fully stable. Clean installer script. Alembic migrations running. Excel export live.
End of Month 2	Employee master + org chart complete. HR department can manage employee records.
End of Month 3	Attendance and leave management live. First integration test with payroll engine.
End of Month 4	Payroll calculation engine (BPJS + PPh 21) complete. Accountant review begins.
End of Month 5	Full payslip PDF + bank CSV. Recruitment module complete. Accountant sign-off.
End of Month 6	Documentation done. Demo environment ready. First external client contract signed.

4. Technical Debt to Clear Before Selling

These items will not crash the app at Garuda — but they will create problems in a client's server room or create legal/security liability.

MUST	Replace <code>_ensure_incremental_schema()</code> ad-hoc ALTER TABLE is unmanageable across multiple client installs	Alembic migrations (Month 1)
MUST	CORS locked to wildcard * any domain can make authenticated requests in production	Lock to client's specific domain on install via .env
MUST	SECRET_KEY has hardcoded fallback predictable secret key in dev deployments	Force env var — raise error if not set
HIGH	No rate limiting brute-force login attacks possible, no API abuse protection	slowapi middleware: 5 req/s per IP on /auth/login
HIGH	No HTTP request logging no forensic trail if something goes wrong at client site	structlog middleware: log method, path, status, duration
HIGH	Password reset flow missing admin must manually reset passwords in DB — not acceptable for clients	SMTP-based reset link, 15-minute expiry token
MEDIUM	No email notifications all notifications are in-app only — users miss events when not logged in	SMTP integration with Mailpit for local dev, SMTP2Go for production
LOW	allow_credentials=False with wildcard CORS technically inconsistent — will break if cookies are ever introduced	Align: either enable credentials with explicit origins, or keep both false/wildcard

5. Commercial Packaging & Pricing

On-premise license model. Client owns the software and hosts it on their own server. Revenue comes from the license fee, implementation project, and annual support contract.

STARTER

Lisensi: Rp 75 juta

Implementasi: Rp 30 juta | Support: Rp 15 juta

PROFESSIONAL

Lisensi: Rp 150 juta

Implementasi: Rp 50 juta | Support: Rp 30 juta

ENTERPRISE

Lisensi: Rp 250 juta

Implementasi: Rp 75 juta | Support: Rp 50 juta

Customization di luar paket: Rp 2–3 juta/jam. Training tambahan: Rp 5 juta/hari on-site.

6. Competitive Positioning

SAP Business One	Rp 500 juta+ license, 6-month implementation, requires certified SAP consultant	10x cheaper, 4-week implementation, no consultant required
Odoo	Too generic — HRIS is shallow, construction modules are third-party add-ons	Construction-specific: WBS, cost codes, AR drives budget ceiling
ACCURATE / Krishand	Desktop-only, no approval workflows, no mobile, no project cost tracking	Modern web, mobile-ready, full 5-step approval matrix, immutable audit log
Spreadsheets + manual	No audit trail, errors not caught, no role separation, version conflicts	Full lifecycle with hard-lock, role-based access, real-time dashboard

7. Revenue Projections

Conservative projections based on on-premise sales cycle (3–4 months close time). Year 1 target: 3 clients. Support contracts compound from Year 2.

New clients	3	5	8
Avg license revenue / client	Rp 175 juta	Rp 200 juta	Rp 225 juta
Avg implementation / client	Rp 55 juta	Rp 60 juta	Rp 65 juta
New client revenue	Rp 690 juta	Rp 1,3 M	Rp 2,3 M
Annual support (existing clients)	—	Rp 90 juta	Rp 240 juta
Total Revenue	Rp 690 juta	Rp 1,4 M	Rp 2,5 M

Breakeven: 2 clients at Professional tier fully cover the development investment. Client 3 onward is net profit. Annual support contracts become material revenue in Year 2.

8. Risks & Mitigations

HIGH	PPh 21 calculation error	
	Impact: Incorrect tax creates liability for client. A tax dispute costs more than the license fee.	Mitigation: Engage a certified accountant (Brevet A/B) to validate the tax engine before shipping. Add a disclaimer in the license that client is responsible for reviewing output.
HIGH	Garuda as sole reference	
	Impact: Potential clients want to see another live installation. One reference is thin.	Mitigation: Target a second friendly company (close relationship, discounted deal) as reference client before the first commercial sale. Offer a 'Beta Partner' deal at cost.
MEDIUM	Construction market is small	
	Impact: Indonesia has ~3,000 mid-size construction companies. Niche market.	Mitigation: Start with construction, then expand to O&M, EPC, mining. The cost control module works for any project-based business. HRIS is universal.
MEDIUM	Payroll complexity underestimated	
	Impact: Indonesian payroll has many edge cases (TK0/K3 PTKP, multi-job, bonus treatment).	Mitigation: Build a comprehensive test suite with 50+ payroll scenarios covering all PTKP statuses, overtime combinations, and THR edge cases before release.
LOW	Client data security on-premise	
	Impact: If client's server is compromised, we may be blamed.	Mitigation: License agreement clearly states client is responsible for server security. Provide a hardening guide in the admin documentation.
LOW	Product name tied to Garuda	
	Impact: Clients may not want software named after a competitor's company.	Mitigation: Rebrand before Month 6 docs are printed. All config uses COMPANY_NAME env var — renaming is a one-line change in the codebase.

9. Immediate Next Steps

Actions to take this week to start the commercialization clock.

1	Wire Excel Export on Reports Effort: 1 day	The Export button exists with no onClick handler. This is the first thing every potential client will try in a demo. Fix it this week.
2	Set up Alembic Effort: 2 days	Run: alembic init alembic. Create the first migration from the current schema. This is foundational — everything else depends on it. Do not start HRIS development before Alembic is running.
3	Build Company Config table Effort: 1 day	Model: CompanyConfig(name, logo_path, npwp, address, city, phone). Admin panel to edit. All PDF templates read from this table.
4	Write the HRIS data model Effort: 1 day (design only)	Sketch the full ER diagram: Employee, Department, JobGrade, AttendanceRecord, LeaveRequest, LeaveBalance, PayrollRun, PayrollLine. Lock the relationships before writing any code.
5	Identify Beta Partner client Effort: Ongoing	Find one friendly construction company willing to be a Beta Partner — discounted or free in exchange for feedback and reference. Start conversations now so they're ready when the product is (Month 6).
6	Lock product name Effort: This week	The name affects all documentation, domain, and branding. Decide before Month 6 — but note it now so it doesn't get forgotten.

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